LECTURE NOTES



1. FUNDMENTALS OF TESTING

1.1 What is testing?

- Money
- Time
- Business reputation
- Injury or Death

* Testing is a set of activities
* Misconception:
- Testing is a process (7 steps)
- Dynamic & Static
- Validation & Verification

1.1.1 Test Objectives

- Finding defects & failures
- Prevent failures
- Validating user needs
- Verifying requirements

1.1.2 Testing & Debugging

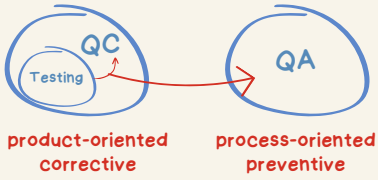
development → defects → testing → defect report → debugging → retesting

1. Reproduce the failure
2. Diagnosis
3. Fix the cause

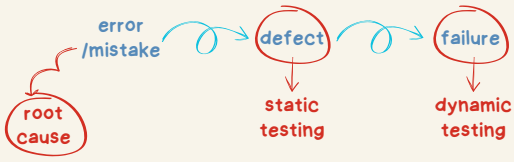
1.2 Why is testing necessary?

- cost-effective
- evaluates quality at different stages of the lifecycles

1.2.2 Testing & Quality Assurance



1.2.3 Errors, defects, failures & root cause



1.3 Testing principles

1. Tesing shows presence of defects not their absence



2. Exhaustive testing is impossible
3. Early testing saves time and money
4. Defects cluster together 80/20



5. Tests wear out



6. Testing is context dependent
7. Absence-of-defects fallacy



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LECTURE NOTES



1

1.4 Test activities, Testware & Test roles

Activities	Testware	Test roles
1. Planning	Test plan, Test schedule, Risk register	Test Manager
2. Monitor & Control	Test progress report	Test Manager
3. Test Analysis what to test	Test conditions (prioritized)	Tester
4. Test Design how to test	Test cases	Tester
5. Implementation are we ready?	Procedures, test suites	Tester
6. Execution	Defect report	Tester
7. Completion	Test completion report	Test Manager

1.4.5 roles of testing

2

1.4.4 Bidirectional Traceability

Test basis ⟷ Testware

Requirements
UI
Code
→ Testing

SRS (product backlog) | Test cases | Defect

1.1
1.2
1.3 xxx
1.4

1. x
2. x
3. x

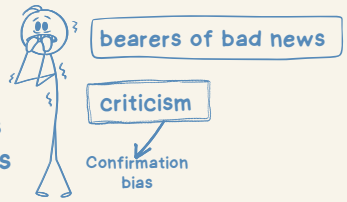
1.x

3

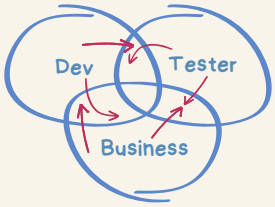
1.5 Essential skills:

1.5.1 General skills

- Test knowledge
- Curiosity
- Attention to details
- Communication skills
- Analytical thinking
- Technical knowledge
- Domain knowledge



1.5.2 whole team approach

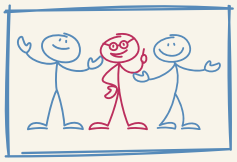


business:
- Acceptance tests

developer:
- Automation
- Test strategy

1.5.3 Independence of testers

4



whole-team approach ✓

less independece
- ✓ communication
- ✓ relationship
- ✓ developer is responsible for quality
- ✗ subjective
- ✗ types of failures

More independece
- ✓ objective
- ✓ different types of failures
- ✗ Isolation
- ✗ Communication
- ✗ Conflicts
- ✗ Developers lose sense of responsibility for quality

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LECTURE NOTES



2. SOFTWARE DEVELOPMENT LIFECYCLE MODELS (SDLC)

Sequential
- waterfall
- v-model

Iterative
- spiral
- prototyping

Incremental
- unified process

Agile lifecycle:
- kanban
- extreme programing xP
- scrum

Development Approaches:

TDD (how)	ATDD (why)	BDD
code -focused write tests write codes refactor	acceptance criteria write tests code ATDD TDD	Given When Then As a user I want to login to access the app A.C Given When Then

Devops & Testing

Development & operations

- Treat both equally

Devops

Dev ∞ Ops

Shift-left ⟷ early testing
- Review requirements
- Review code
- Write test cases before code
- Static analysis

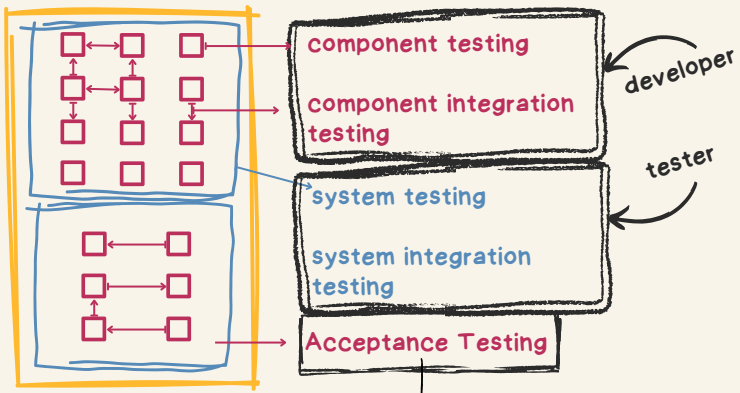
Retrospectives:
- What was successful ✓
- What was not successful ✗
- How to incorporate the improvements

2.2 Test levels & Test types

Test levels:

Mobile Version

Web Version



- users
- product owner
- business analyst

Test types:

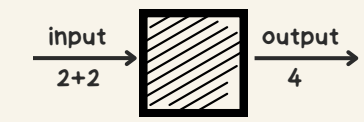
Functional testing:

what the test object does (yes / no)
- completness
- correctness
- appropriateness

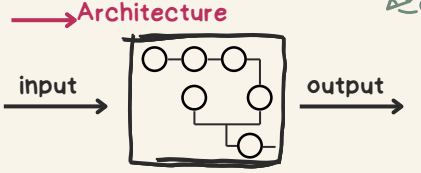
Non-functional testing:

How well the system behaves
- performance
- usability
- security

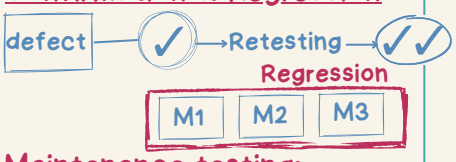
Black-box testing:



White-box testing:



Confirmation & Regression:



Maintenance testing:
- planned releases
- unplanned releases (hot fixes)

Triggers for maintenance:
- modification
- upgrades or migration
- retirement

LECTURE NOTES



3. STATIC TESTING

- Reviews: any work product
- Static Analysis

Static testing finds defects
Dynamic testing finds failure

Code → IDE → warning / errors

Review process:

- Planning → identify work product & scope
- Review initiation → prepare for review
- Individual review → create anomalies
- Communication & Analysis → review anomalies
- Fixing & Reporting

Review types:

- informal review — detect anomalies
- walkthrough — leader = author
- Technical review — technically qualified reviewers; led by moderator
- Inspection — find maximum number of anomalies

Roles & Responsibilities :- Reviews:

Manager	Author	Moderator	Scribe	Reviewer	Review leader
- Decides - Provides resources	- Create work product - Fix work product	- Mediation -Time Management	- Records decisions	- Review work product	take overall responsibility

Success factor for reviews:

Small chunk
- concentration → reviewers
- provide feedback → author

LECTURE NOTES



4. TEST ANALYSIS AND DESIGN

small number of test cases → Achieve → highest coverage possible

Test techniques

1. Black box techniques (specification-based)

1) Equivalence partitioning (EP)

15 ✗ | ✓ 5

10

2) Boundary-value analysis:

10 11 *two point BVA

10 11

9 10 11 10 11 12

9, 10, 11, 12 *three point BVA

* 10 , 11

No discount — Discount

3) Decision-table testing:

1. Previous experience Y/N
2. Are you certified Y/N

4 table rules

Conditions	1	2	3	4
Previous experience	Y	Y	N	N
Certified	Y	N	Y	N
Actions				
Get discount			✓	✓
Go to basic course		✓		✓

2 — Conditions

Action

2. White-box techniques

✓ Statement coverage:

1. 1. 3. 4. 5.

100% SC

if X > 5 → 7

else ✗

+ Branch coverage

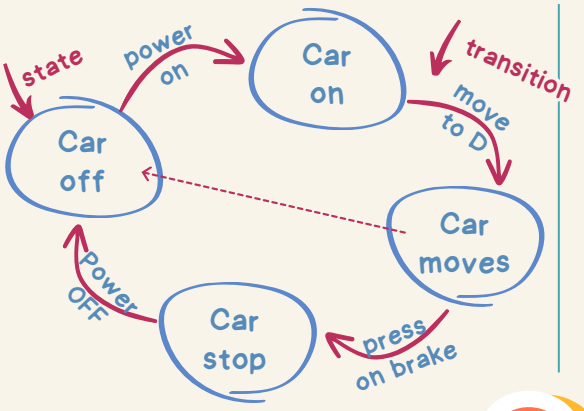
1. 1. 3. 4. 5.

if X > 5 → 7 / 3

else ✗

DC SC

4) State-transition testing:



* all-states coverage (4 states)
* all valid-transitions coverage (4 transitions)
* all transitions coverage

3. Experience-based techniques

1) Error guessing:

Fault attack

2) Exploratory testing:

Analysis → Design → Execute

Test charter

Session-based SBTM

3) Checklist-based testing:

Are all redirects working?

collaboration-based test approaches:

collaborative user-story writing:

3C

card — conversation — confirmation

how software will be used

acceptance criteria

senario-oriented: Given When Then

rule-oriented

ATDD Acceptance test-driven development

user story

Acceptance tests

might be automated

write code using TDD

1. positive tests
2. negative tests
3. non-functional tests

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LECTURE NOTES



5. TEST MANAGEMENT

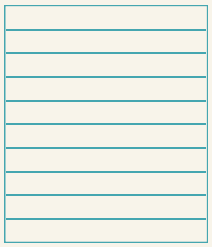
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Test planning → Test plan

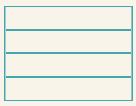
- schedules
- test activities
- communication
- assumptions & cost reviews
- risk register
- budget
- test approches

2

Release planning & iteration planning:



Release ← writing testable user stories



Iteration ← break down user stories into testing tasks

3

Entry criteria & Exit criteria:

Entry criteria (DOR)		Exit criteria (DOD)
Code is checked in	→ Integration testing →	All interfaces tested
Static analysis Done	→	System testing is ready
100% BC	→	No blockers
		Coverage achieved

4

Estimation techniques:

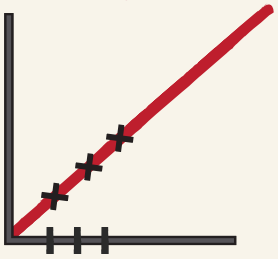
- metrics based
- expert based

1) Estimation Based on Ratios:

✓5 developers → ✓15 developers ✓100 hrs dev ✓10 hrs testing

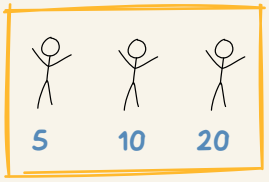
✓2 testers → 6 testers ✓ 20 hrs dev 2 hrs testing

2) Extrapolation:



3) Wideband Delphi: (Planning poker)

experts makes estimations → in isolation



→ 12

4) Three-point estimates

O = 2
M = 5
P = 30

$$E = \frac{O + 4M + P}{6} = \frac{2 + 5 \times 4 + 30}{6} = \frac{52}{6} = 8.667$$

$$SD = \frac{30 - 2}{6} = \frac{28}{6} = 4.67 \pm \quad \text{Estimate} = 8.667 \pm 4.67$$

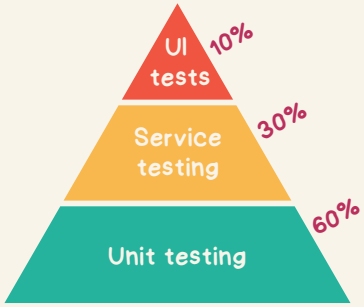
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Test case prioritization

risk -based	coverage -based	requirement -based
high risk → medium risk → low risk	highest additional -coverage	

6

Test pyramid



LECTURE NOTES



1

Testing quadrants:

- Q2 (Functional): - business, - support
- Q3 (exploratory UAT): - business, - critique
- Q1 (Component): - technology, - support
- Q4 (Non-functional tests): - technology, - critique

3

Risks types: product risks, project risks

2

Risk management:

1. Risk identification
2. Risk assessment (likelihood, impact) } risk analysis
3. Risk mitigation } Risk control
4. Risk monitoring

5

Configuration management:

testware (test items)
- uniquely identified
- version controlled
- tracked for changes
- related to other configuration items

4

Monitoring, control & completion

- gathering information
- corrective actions

→ test progress report

→ test completion → test completion report

6

Defect management:

- ID
- Title
- Date
- Test object
- Steps to reproduce
- Expected result
- Actual result
- Severity
- Priority
- Status

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LECTURE NOTES



6. TEST TOOLS

- Management tools
- Static testing tools
- Design & Implementation
- Execution & Coverage tools
- Non-Functional test tools
- DevOps tools
- Collaboration tools
- Scalability & Deployment

1

2

Benefits of Automation	Risks of Automation
- Reducing time - Prevention of human errors - Easier access to information	- Unrealistic expectations - Relying on tool too much - Vendor problems - Open-source tools